

197. Rising Temperature

PLATFORM

Platform: LeetCode

DIFFICULTY

EASY

DATE

Solved: June 4, 2026

Problem

Find all dates where the temperature was **higher than the previous day's temperature**.

Return the `id` of those records.

Execution Flow

1. Join `Weather` with itself.
2. Match each record with the previous day's record using `DATEDIFF`.
3. Compare temperatures.
4. Return ids where today's temperature is higher.

Concepts Used

1. Self Join

```
FROM Weather w1
JOIN Weather w2
```

The same table is joined with itself.

- `w1` → Current day
- `w2` → Previous day

This allows us to compare temperatures between two different rows.

2. Match Consecutive Days

```
ON DATEDIFF(w1.recordDate, w2.recordDate) = 1
```

`DATEDIFF(date1, date2)` returns the number of days between two dates.

Example:

```
DATEDIFF('2015-01-02', '2015-01-01') = 1
```

This condition ensures:

- `w1` is exactly one day after `w2`.

3. Compare Temperatures

```
WHERE w1.temperature > w2.temperature
```

Checks whether today's temperature is greater than yesterday's temperature.

Database

197. Rising Temperature

Solved

Easy

Topics

Companies

SQL Schema

Pandas Schema

Table: Weather

Column Name	Type
id	int
recordDate	date
temperature	int

id is the column with unique values for this table. There are no different rows with the same recordDate. This table contains information about the temperature on a certain day.

Write a solution to find all dates' `id` with higher temperatures compared to its previous dates (yesterday).

Return the result table in **any order**.

The result format is in the following example.

Example 1:

Input:

id	recordDate	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

Output:

id
2
4

Accepted

15 / 15 testcases passed

Sahil Khan submitted at Jun 04, 2026 18:53

Runtime

584 ms Beats 10.96%

Code | MySQL

```
1 # Write your MySQL query statement below
2 SELECT w1.id
3 FROM Weather w1
4 JOIN Weather w2
5 ON DATEDIFF(w1.recordDate, w2.recordDate)=1
6 WHERE w1.temperature>w2.temperature
7
```

More challenges

- 1747. Leetflex Banned Accounts
- 1951. All the Pairs With the Maximum Number of Common Followers
- 3055. Top Percentile Fraud

Write your notes here

Code

MySQL

```
1 # Write your MySQL query statement below
2 SELECT w1.id
3 FROM Weather w1
4 JOIN Weather w2
5 ON DATEDIFF(w1.recordDate, w2.recordDate)=1
6 WHERE w1.temperature>w2.temperature
7
```

Testcase

Test Result

Accepted

Runtime: 64 ms

Case 1

Input

id	recordDate	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

Output

id
2
4